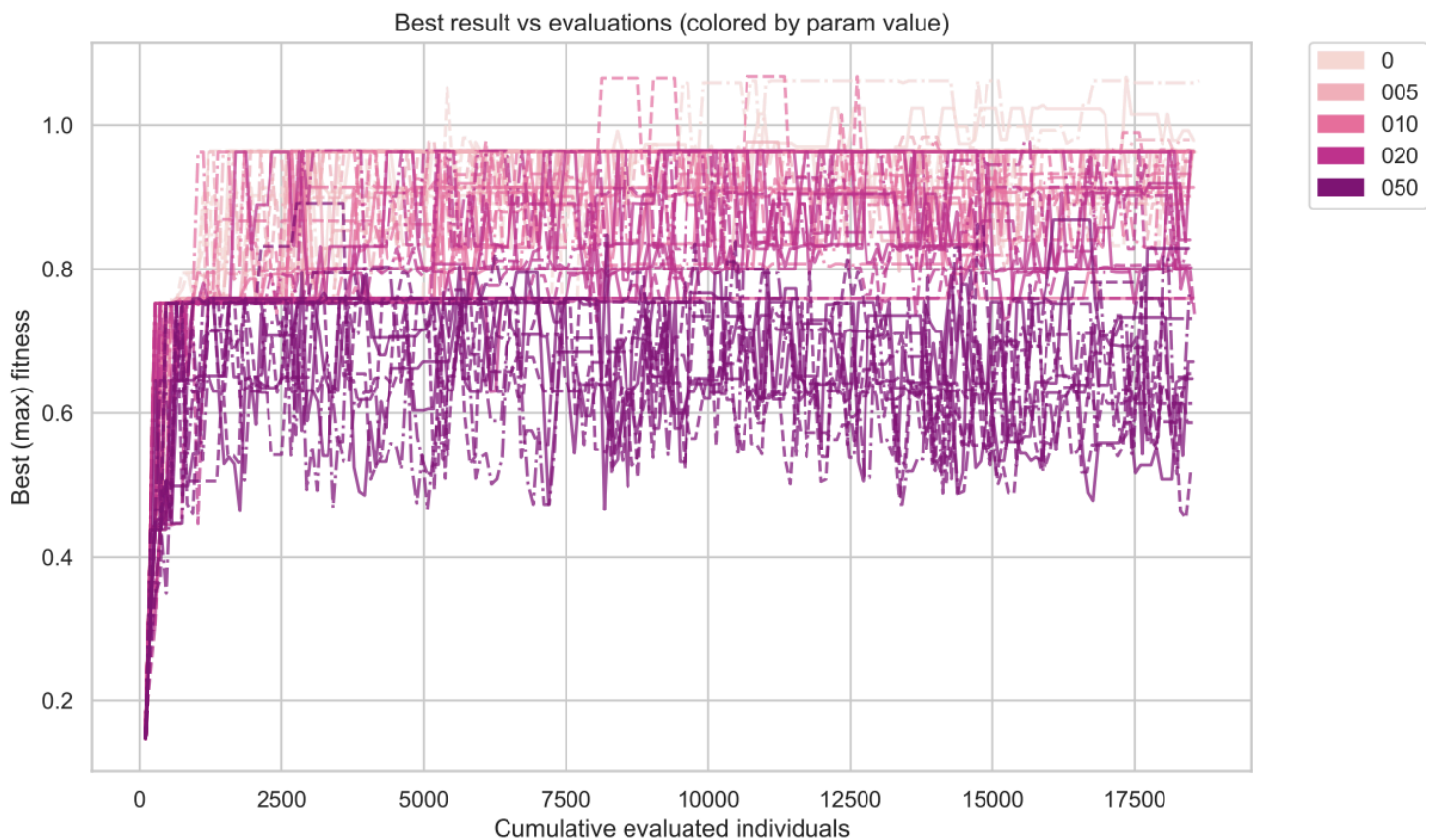


Task 4

The following mutation intensity values were used: 0, 0.05, 0.10, 0.20, 0.5.

Total number of generations was set to 200, with 100 individuals in population, and tournament size 10. 10 experiments were run for each parameter value, so in total 50 experiments, [run in parallel on 24 cores].

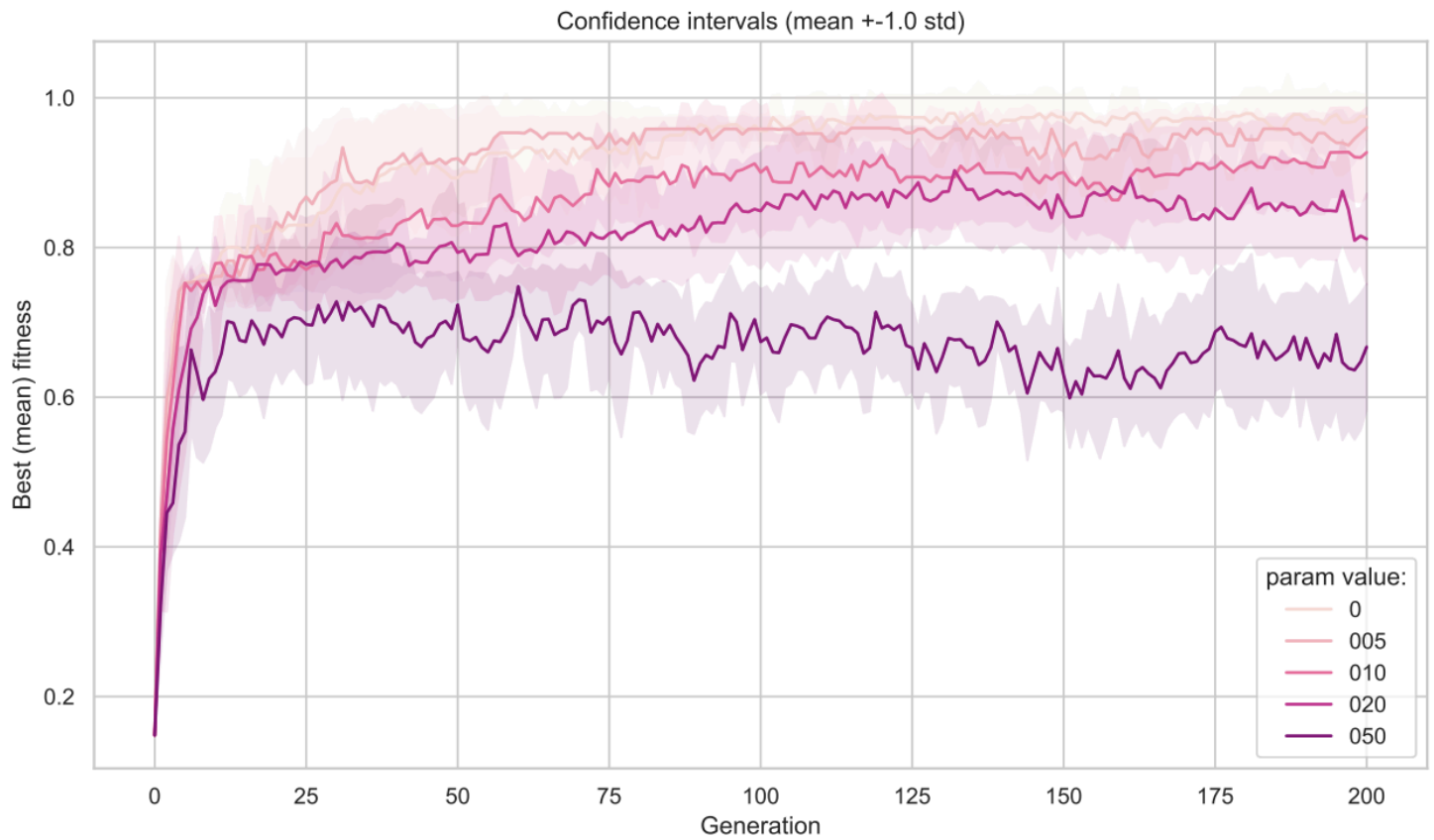
Despite trying several experiment settings (that is playing around tournament size, population size), I was not able to achieve some degree of convergence, as demonstrated on the plot below:



Each curve on the plot above correspond to the best individual in the population's HoF (of size 1).

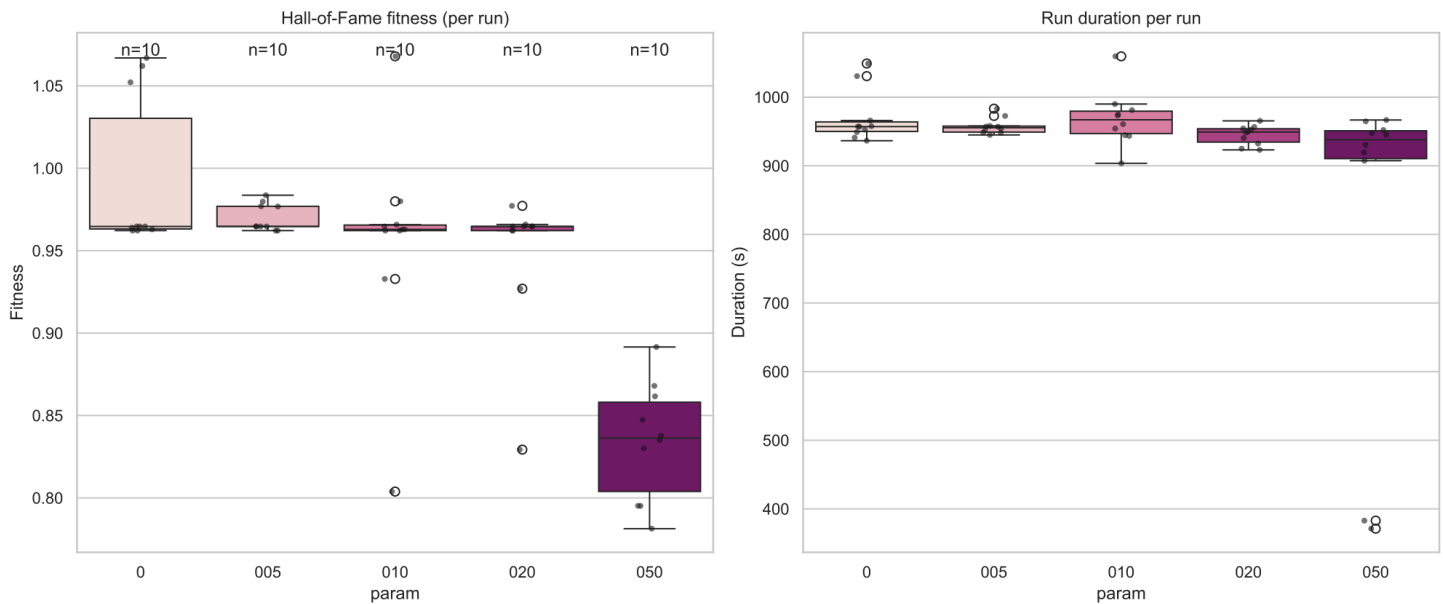
Each curve seems not to converge to any specific value in a strict mathematical sence: all the curve fluctuate below the best achieved value in the series (e.g. 1.06 for series with mutation intensity 0).

The next plot demonstrates averages over generations with 67% confidence limits.



Note: In this plot preference in x-axis was given to generation instead of number of evaluated individuals due to simpler calculations of mean and std from log files

Boxplots of best fitness of HoF representatives, and total time spent on evaluations.



Note: 2 outliers on the most right boxplot can be explained by the fact that all the 50 experiments were scheduled on 24 cores, and the last 2 experiments were run on almost free cpu which reduced overall running time.

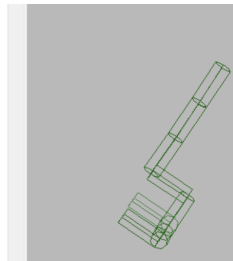
One can conclude based on these plots that most beneficial mutation intensity is between 0 and 0.05, though 0 demonstrates the best results after 100 generations, which sounds quite counterintuitive. That is far away from my predictions of 0.27... or 0.1. At the same time, increasing number of generations could be beneficial in discovering new tendencies.

Since mutation intensity directly affects the size of neighbourhood, it indeed looks surprising that 0 mutation intensity yields the best results.

Basically mutation works as follows: an offspring (one of 2) of 2 selected parents, which is crossed-over with default probability of 0.2, is mutated with probability mutpb (a value 0, or 0.5, ...). Mutation procedure means that a) this offspring is left intact with prob $(1 - \text{mutpb})$, or b) with probability mutpb this offspring is instantly substituted with its neighbor at distance 1. Let's consider the second option: mutation means, for instance, that a long structure suddenly acquires a new limb at random place with random direction or gets some part of it substituted with another limb. If this happens on the "vertebra", with very high probability a high stable creature becomes less tall and more unstable: if only 6 genome letters indicating limb direction are available, then this happens in 4 out of 6 cases.

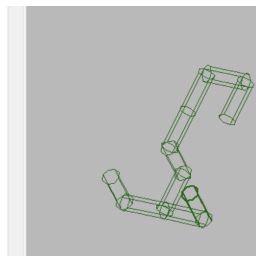
In the following example changing one genome letter immediately changes the form of the creature:

`/*9*/BFLBFRUBUUUD`



-> this creature immediately becomes this ->

`/*9*/BFLBFRUBUURD`



The first naturally emerged creature - though unstable, immediately becomes less tall and more "screwed" (entropy increased, and significant evolution progress seems to roll back).

To summarize, mutation very likely decreases a creature's length and increases its instability.

Now, if we think about mutation intensity as of "radius" of neighbourhood, it becomes clear that random selection of any of neighbour in the large neighbourhood acts similarly to random walk, since the further the neighbour, the less connection to parent's well performing genome. Algorithm basically "jumps" to a likely distant position without researching an "adjacent ring".

Now I try to explain why crossover only can be the driver of the evolution process in these experiments.

Crossover works as splitting genome at random place (with some probability set to 0.2). Let's assume we have 2 parents with either tall or stable bodies. Tall structures usually emerge as a sequence of U in the genome, and any such sequence represents the most compact way of representing a tall "vertebra" (compared to UDUD...). Splitting a creature's genome at any random place most likely catches long U-sequences. Thus they are easy to inherit during CX. As for stabilizers, anything at the "left"/"bottom" end of creature with massive number of letters can serve as a future substrate for the "vertebra" (spine) caught from another parent. Thus combining both stable/"fat" parents with tall and possibly not so stable ones really boosts evolution. I want to point out that both such kinds of parents should be often met in the population since both has more random genome embodying more entropy and uniformity than their offspring with nicer combined properties.

Conclusions:

1. Lack of smooth "plateau" at the end of 200 generations suggests that population have not reached a stationary state. There are 2 possible explanations: the operators used have very disruptive nature (as was pointed out for mutation operator), or there is a potential for improvement, since there is no theoretical max bound on the maximum vertical position of mass center, and current experiment configuration can theoretically yield much better results.
2. Evidence supports the locality principle: offsprings should not very different from their parents. High mutation intensity acts like random walk, which is absolutely inefficient in reaching high-quality peaks. Setting mutation intensity to 0 means we entirely rely on crossover, which preserves building blocks.
3. Building-block hypothesis: it seems that in this particular experiment (i.e. series of experiments) crossover efficiently constructs organisms using long U-sequences, as described above, and we likely would be able to observe even longer U-schematas and better results if the experiment were run longer.